Practice Exam: Genetics (Version 2)

Name: _____

Question type	Points	Time budget
24 multiple choice	3 each (72)	32 min
6 short answer	2 each (12)	6 min
1 essay	25	12 min

Each question tests the same concept as the matching number on Sample Exam 4, but from a different angle. Closed book; no calculator needed.

Multiple Choice

1. A cell has **2N chromosomes and 4X DNA content**. It could be in:

A. G1
B. G2
C. The end of meiosis I
D. A gamete
E. The end of mitosis

2. An organism has **2N = 6**. You see a cell with **3 X-shaped chromosomes, all different sizes**. This cell is most likely in:

A. G2 of mitosis
B. Prophase I
C. Metaphase II
D. Telophase of mitosis
E. G1

3. Which event happens in **meiosis but NOT mitosis**?

A. Sister chromatids separate
B. Chromosomes condense
C. Homologous chromosomes pair and cross over
D. A spindle forms
E. Cytokinesis

4. A person is **AaBb** with the genes on **different chromosomes**. Which observation is evidence of independent assortment?

A. Gametes AB, Ab, aB, and ab are made in equal proportions
B. Only AB and ab gametes are made
C. Each gamete gets only one A-gene allele
D. All gametes are identical

5. In a testcross **Aa x aa**, the expected **genotypic and phenotypic** ratios are:

A. 1:2:1 and 3:1
B. 1:1 and 1:1
C. 1:1 and 3:1
D. 3:1 and 3:1

6. How many genetically different gametes can an **AaBbCC** individual make (unlinked genes)?

A. 2
B. 4
C. 6
D. 8
E. 16

7. Colorblindness is X-linked recessive. A **colorblind woman** has children with a man with **normal vision**. Expect:

A. All sons and all daughters colorblind
B. All sons colorblind, no daughters colorblind
C. Half of sons and half of daughters colorblind
D. No sons colorblind, all daughters colorblind
E. Half of sons colorblind, no daughters colorblind

8. Two **unaffected** parents have an **affected daughter** and an unaffected son. The trait is most likely:

A. Autosomal dominant
B. Autosomal recessive
C. X-linked recessive
D. Any of these

9. A dihybrid F1 (from **AABB x aabb**) is testcrossed. Offspring: **AaBb 410, aabb 390, Aabb 105, aaBb 95.** These results suggest the genes are:

A. Unlinked, on different chromosomes
B. Linked, with about 20% recombination
C. Completely linked, with no crossing over
D. Far apart on the same chromosome (about 50% recombination)

10. Which F2 ratio from a dihybrid cross would suggest **gene-by-gene interaction**?

A. 3:1
B. 1:2:1
C. 9:3:3:1
D. 9:7
E. 1:1:1:1

11. A person **heterozygous** for a recessive disease allele has a **normal phenotype**. What is most likely true?

A. The recessive allele is never transcribed
B. Both alleles may be transcribed and translated, but the normal protein masks the effect of the recessive one
C. The recessive allele was deleted from the genome
D. The gene is not present in somatic cells

12. A gene has **5 different alleles** in a population. What is the **maximum** number of alleles of this gene one diploid person can have?

A. 1
B. 2
C. 5
D. 10

13. A double-stranded DNA sample is **22% guanine**. What percent is **adenine**?

A. 22%
B. 28%
C. 44%
D. 56%
E. 78%

14. If DNA replication were **conservative**, after **one** round of replication in ^{14}N (starting from ^{15}N DNA), you would see:

A. All intermediate DNA
B. One heavy band and one light band
C. One heavy band and one intermediate band
D. All light DNA

15. A primer **5'-AGC-3'** is paired with the template **3'-TCGTAC-5'**. What nucleotide will DNA polymerase add **next**?

A. Adenine
B. Thymine
C. Guanine
D. Cytosine
E. None; polymerase can't extend this primer

16. Which flow of information is **NOT** possible according to the Central Dogma?

A. DNA to DNA
B. DNA to RNA
C. RNA to DNA
D. RNA to protein
E. Protein to RNA

17. A mutation prevents an **activator transcription factor** from binding near the promoter of gene X. Most likely result:

A. Protein X is made but is non-functional
B. Less mRNA is made, so there is less protein X, but the protein is normal
C. mRNA is made but can't be translated
D. Gene X becomes heterozygous

18. The **coding strand** of a gene reads **5'-ATGCCA-3'**. The mRNA is:

A. 5'-UACGGU-3'
B. 5'-AUGCCA-3'
C. 5'-UGGCAU-3'
D. 3'-AUGCCA-5'

19. Which is found **only** in prokaryotic genomes?

A. Introns
B. Operons
C. Chromatin
D. Multiple linear chromosomes
E. Transposable elements

20. Rice has about 37,000 genes; humans have about 19,000. The best explanation for why humans aren't "simpler" than rice:

A. Rice is actually more complex than humans
B. Gene number doesn't predict complexity; for example, alternative splicing lets one human gene make several proteins
C. Humans have fewer chromosomes
D. Most rice genes are non-coding

21. How many amino acids are in the polypeptide translated from this mRNA?
5'-GGAUGUUUGCAUAAGC-3' (use the codon table on the last page)

A. 2
B. 3
C. 4
D. 5

22. Which mutation is **most likely to have NO effect** on the protein?

A. Deleting one nucleotide in codon 2
B. A base change that creates a stop codon in codon 5
C. A third-position base change that doesn't change the amino acid
D. A missense change in the enzyme's active site

23. Which is true of **both** prokaryotes and eukaryotes?

A. 5' cap
B. RNA splicing
C. AUG start codon
D. Coupled transcription and translation
E. Chromatin

24. Which mutation would exome sequencing **miss**?

A. A nonsense mutation in exon 3
B. A frameshift in exon 1
C. A mutation in an enhancer 10,000 bp upstream of the gene
D. A missense mutation in the last exon

Short Answer

Define each term in a **single sentence** (2 pts each).

25. Sister chromatids
26. Co-dominance
27. Transcription factor
28. Crossing over
29. Linkage
30. Semi-conservative replication

Essay (25 pts)

A couple are **both carriers** of cystic fibrosis (CF), which is **autosomal recessive** (CFTR gene on chromosome 7).

1. Explain how **meiosis** produces each parent's gametes, and give the probability that a child is **affected, a carrier, and unaffected (not a carrier)**. Name the Mendelian law involved and the meiotic stage responsible.

2. The most common CF mutation **deletes exactly 3 nucleotides**. Explain why this is **not a frameshift**, and how it could still cause disease.

3. Describe how the **CFTR protein is made from the gene** in a healthy person, including **where** each step happens and the **key players** at each step.

Essay (continued)

Codon Table

1st \ 2nd	U	C	A	G
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA STOP UAG STOP	UGU Cys UGC Cys UGA STOP UGG Trp
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg
A	AUU Ile AUC Ile AUA Ile AUG Met (Start)	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly